LOCAL AND GLOBAL STRUCTURE OF AGRAWAL'S CONGRUENCE AT $r = 5$: GOLDEN KUMMER DEFECTS, DENSITY, AND FINITE FIBRES

[AUTHOR LINE TO BE DECIDED]

ABSTRACT. Let r be prime, $r \nmid n$, and consider Agrawal's congruence

$$(X-1)^n \equiv X^n - 1 \pmod{n, X^r - 1}.$$

Agrawal's conjecture predicts that the congruence forces n to be prime or $n^2 \equiv 1 \pmod{r}$. We study the case $r = 5$. For a prime factor $p \mid n$ we classify the local exponent set through discrete logarithms of the cyclotomic units $\zeta_5^a - 1$. The second cyclotomic moment is governed by the golden unit $\varepsilon = (1+\sqrt{5})/2$, and the same unit controls a minimal Kummer field for the quartic tail anomaly. We obtain exact fixed-level densities, finite-product disjointness, an unconditional density-zero theorem for the possible order-four branch, and, under Dedekind GRH for explicit Kummer composita, the quantitative bound

$$N_H(X) \ll \frac{X}{(\log X)^2}.$$

A separate Artin-Hooley type calculation gives, under GRH, the exact relative density

$$1 - \prod_{\ell \neq 2,5} \left(1 - \frac{1}{\ell(\ell-1)}\right) = 1 - \frac{40\mathcal{A}}{19}$$

for quartic tail anomalies.

For squarefree n, assuming the order-four hypothesis H4, we reduce every counterexample to an explicit generalized CRT system followed by a finite last-prime fibre. The two-factor and all even-factor cases are excluded. Seven three-factor fibres are certified empty without a bound on the last prime, and a larger finite box is excluded by exact computation. We prove that every squarefree counterexample is a Fermat pseudoprime to base 5, placing the second global obstruction inside a classical counting problem. The uniform bound of Li–Pomerance then gives

$$\sum_s M_s(X) \leq X^{1 - \log\log\log X/(2\log\log X)}$$

for all sufficiently large X. We also audit three natural attacks on the two remaining walls: the BHV threshold does not apply to the fourth-order fibre sequences; Iwasawa invariants do not decide the moving-prime condition; and the natural fixed 2-adic logarithmic forms have uniformly bounded valuation. Two obstacles therefore remain: H4 and the global vacuity of the union of finite fibres.

CONTENTS

Version 17, 24 July 2026. This is an audit-oriented research draft. Every computational statement is assigned a finite domain or a certificate. Statements labelled conditional are not promoted to unconditional theorems. An elementary core of eight modules and 45 named declarations is formalized in Lean 4 with Mathlib, without `sorry`, `admit`, or added axioms.

1. Introduction

Agrawal's conjecture asks whether a single polynomial congruence already contains enough information to certify primality, apart from the unavoidable escape $n^2 \equiv 1 \pmod r$. In the form used here it states:

Conjecture 1.1 (Agrawal)**.** *Let r be prime and $r \nmid n$. If*

$$(X-1)^n \equiv X^n - 1 \pmod{n, X^r - 1},$$

then n is prime or $n^2 \equiv 1 \pmod r$.

The local algebra has a simple entry point. For $p \neq r$, let

$$A_{p,r} = \mathbb{F}_p[X]/(\Phi_r(X)), \qquad \zeta = X \bmod \Phi_r,$$

and define

$$S(p,r) = \{m \geq 1 : (\zeta - 1)^m = \zeta^m - 1 \text{ in } A_{p,r}\}.$$

When n is squarefree, the global congruence is equivalent to

$$\frac{n}{p} \in S(p,r) \qquad (p \mid n).$$

Thus the problem separates into a local classification of $S(p,r)$ and a global compatibility problem among the exponents n/p.

The case $r = 5$ is the first case in which the local image can carry a genuine order-four ambiguity. It is also the smallest case in which the real quadratic subfield $\mathbb{Q}(\sqrt{5})$ and its fundamental unit enter canonically.

Contributions and status. The results fall into four layers.

1. A local theory of $S(p,5)$, including subgroup structure, cyclotomic moments, the golden-unit formula, exact order decompositions, and Kummer interpretations.
2. An autonomous formulation of the possible order-four branch H4, its reduction to a support problem for Fibonacci-Lucas gcds, density zero unconditionally, and an $X/(\log X)^2$ upper bound under GRH.
3. A squarefree reduction theorem under H4. Counterexamples correspond exactly to survivors in explicit finite fibres indexed by compatible tuples. This is a reduction theorem, not a proof that all fibres are empty.

(4) Certified computations: three Fermat-positive J-profiles, no H-profile up to 10^9, a certified H_n census to $n = 100000$, seven empty three-factor fibres, and a universal-level factor cache.

No claim in this draft implies the full conjecture. H4 and global fibre vacuity remain open.

2. The local exponent group

Proposition 2.1 (Frobenius and multiplication). *The set $S(p, r)$ contains 1 and p, is closed under multiplication, and is stable under multiplication by p. Moreover, every $m \in S(p, r)$ is coprime to r.*

Proof. The Frobenius identity gives

$$((\zeta - 1)^m)^p = (\zeta^m - 1)^p = \zeta^{pm} - 1.$$

For multiplication, apply the endomorphism $\zeta \mapsto \zeta^m$ to the relation for a second exponent. Coprimality follows because $\zeta - 1$ is a unit and $\zeta^r = 1$. $\square$

After quotienting by the natural period, the finite cancellative semigroup is a group. Write $\operatorname{im}_5 S(p, 5)$ for its image in $(\mathbb{Z}/5\mathbb{Z})^\times$.

Choose a splitting field containing ζ_5, a generator γ of its multiplicative group, and write

$$\zeta_5^a - 1 = \gamma^{e_a}.$$

Then

$$m \in S(p, 5) \iff me_a \equiv e_{am} \pmod{\#\langle\gamma\rangle} \quad (a = 1, 2, 3, 4).$$

This turns the local congruence into a linear system in discrete logarithms.

3. Cyclotomic moments and the golden unit

For $j \in \mathbb{Z}/4\mathbb{Z}$, the cyclotomic moments are weighted transforms of the four logarithms e_a. Their exact normalization is chosen so that nonvanishing moments cut out the possible residue classes of m mod 5.

Theorem 3.1 (Moment obstruction). *There is an explicitly defined subgroup $T(p, 5) \leq (\mathbb{Z}/5\mathbb{Z})^\times$, determined by the nonzero cyclotomic moments, such that*

$$\operatorname{im}_5 S(p, 5) \subseteq T(p, 5).$$

The moments satisfy Frobenius, parity, head, and tail identities.

The key special feature of $r = 5$ is the following identity.

Theorem 3.2 (Golden moment). *Let*

$$\varepsilon = \frac{1 + \sqrt{5}}{2}.$$

For $p \equiv 1 \pmod 5$, with compatible quintic index normalization,

$$M_2 = 3 \operatorname{ind}_5(\varepsilon).$$

Together with $M_0 = \operatorname{ind}_5(5)$, the pair (M_0, M_2) gives invertible coordinates on the two Kummer directions generated by 5 and ε.

The integral cyclotomic identity behind the later entanglement calculations is

$$\boxed{5 = \zeta_5^3 \varepsilon^2 (\zeta_5 - 1)^4.}$$

Here $\varepsilon = -(\zeta_5^2 + \zeta_5^3)$ inside $\mathbb{Z}[\zeta_5]$.

4. Local classification and H4

The local residue degree is

$$f_p = \mathrm{ord}_5(p) \in \{1, 2, 4\}.$$

It must not be confused with the global number of prime factors.

Theorem 4.1 (Quartic rigidity)**.** *If $f_p = 4$, equivalently $p \equiv \pm 2 \pmod 5$, then*

$$S(p, 5) = \langle p \rangle$$

in the natural exponent group.

If $f_p = 1$ or 2, the only possible non-Frobenius residue has order four. This motivates:

Hypothesis 4.2 (H4)**.** *No prime $p \neq 2, 5$ in either split branch $p \equiv \pm 1 \pmod 5$ admits the order-four local transport symmetry.*

H4 has several equivalent forms. Put

$$A_n = \begin{cases} L_n, & n \text{ even}, \\ F_n, & n \text{ odd}, \end{cases} \qquad H_n = \gcd(A_n, 5^{n-1} + 1), \qquad n \equiv 4 \pmod 5.$$

Theorem 4.3 (Support formulation)**.** *The following are equivalent.*

1. *H4.*
2. *Every odd prime divisor $p \neq 5$ of every H_n, $n \equiv 4 \pmod 5$, is inert in $\mathbb{Q}(\sqrt{5})$.*
3. *No split prime satisfies the exact order profile*

$$R_p = 2r, \qquad E_p = 2s, \qquad \gcd(r, s) = 1, \qquad r + s \text{ odd}, \qquad 5 \nmid rs,$$

where $R_p = \mathrm{ord}_p(5)$ and $E_p = \mathrm{ord}_p(\varepsilon^2)$.

Theorem 4.4 (Inert support)**.** *Every inert prime $p \neq 2, 5$ divides H_n for infinitely many $n \equiv 4 \pmod 5$. If $p \equiv 2 \pmod 5$, then the canonical occurrence is*

$$p \mid H_{(p+1)/2}.$$

Thus H4 asserts an equality of supports:

$$\bigcup_{n \equiv 4(5)} \mathrm{Supp}(H_n) = \{p : (5/p) = -1\},$$

away from 2 and 5.

5. Density of H4 candidates

For an odd prime $\ell \neq 5$, let B_ℓ be the event

$$\ell \mid p - 1, \qquad 5 \notin (\mathbb{F}_p^\times)^\ell, \qquad \varepsilon \notin (\mathbb{F}_p^\times)^\ell.$$

A split H-profile must avoid every B_ℓ. The fixed-level relative density is

$$d_\ell = \frac{\ell - 1}{\ell^2}.$$

Theorem 5.1 (Finite-product disjointness)**.** *For every finite set of odd primes $\ell \neq 5$, the corresponding Kummer fields have the product disjointness needed to multiply the densities of the events B_ℓ.*

Corollary 5.2 (Density zero)**.** *The set of H4 candidates has natural density zero.*

Proof. For fixed Y, finite Chebotarev gives an upper density bounded by

$$C \prod_{\substack{\ell \le Y \\ \ell \ne 2,5}} (1 - d_\ell).$$

The product is asymptotic to a constant divided by $\log Y$. Let Y tend to infinity after the prime-counting limit. $\square$

Theorem 5.3 (Quantitative H4 compression under GRH)**.** *Assume Dedekind GRH for the explicit squarefree Kummer composita used by the joint sieve. For each split branch,*

$$\boxed{N_{H,i}(X) \ll \frac{X}{(\log X)^2}.}$$

Proof outline. For squarefree d, the union of Chebotarev classes defining $\cap_{\ell | d} B_\ell$ has density $g(d)$. The class-sensitive normalization preserves this density after summing disjoint classes, and

$$\delta_i g(d)[K_{i,d} : \mathbb{Q}] = \varphi(d)^2.$$

Consequently the GRH error is

$$r_i(d; X) \ll \varphi(d)^2 \sqrt{X} \log(Xd).$$

The quadratic expansion of the Selberg upper-bound weights contributes

$$\sum_{d < z^2} 3^{\omega(d)} |r_i(d; X)|.$$

Taking $z = X^{1/14}$ gives an error $O(X^{13/14} \log^3 X)$, while the sieve product contributes $X/(\log X)^2$. $\square$

6. Quartic tail anomalies

Let $p \equiv \pm 2 \pmod 5$, and let E_ℓ denote the failure at the odd Kummer level $\ell \mid p + 1$. The elimination identity is

$$(\zeta_5 - 1)^{(p-1)(p^2+1)} = \varepsilon^2.$$

It reduces the event to a minimal field

$$M_\ell = \mathbb{Q}(\sqrt{5}, \mu_\ell, \varepsilon^{1/\ell})$$

of degree $2\ell(\ell - 1)$. The event is represented by a central singleton conjugacy class.

Theorem 6.1 (Tail density under GRH)**.** *Assume Dedekind GRH for the fields M_ℓ. The relative density, among the quartic primes, of primes with at least one tail anomaly is*

$$\Delta = 1 - \prod_{\ell \ne 2,5} \left(1 - \frac{1}{\ell(\ell-1)}\right) = 1 - \frac{40\mathcal{A}}{19} = 0.212724602906942\ldots,$$

where $\mathcal{A}$ is Artin's constant.

The proof combines finite-product disjointness, a class-sensitive Chebotarev estimate up to a root-log threshold, Brun-Titchmarsh in the remaining logarithmic band, and an unconditional large-level elimination.

7. Squarefree global reduction

Let n be squarefree and write $s = \omega(n)$.

Theorem 7.1 (Structure modulo 5)**.** *If a squarefree composite* n *satisfies Agrawal's congruence at* $r = 5$ *and* $n^2 \not\equiv 1 \pmod 5$*, then an even value of* s *forces an order-four local symmetry. Under H4,*

$$s \geq 3 \textit{ is odd}, \qquad p \mid n \Longrightarrow p \equiv \pm 2 \pmod 5.$$

In particular, under H4 the two-factor case and every even-factor case are excluded.

Let

$$n = p_1 \cdots p_{s-1} q = Pq, \qquad \nu = n \bmod 5 \in \{2, 3\}.$$

For every prefix prime define $j_i \in \{0, 2\}$ by

$$j_i = 0 \iff p_i \equiv \nu \pmod 5,$$

and let T_i be the order of $\zeta_5 - 1$ in $\mathbb{F}_{p_i^4}^\times$.

Theorem 7.2 (Generalized CRT machine)**.** *The first* $s - 1$ *local rows are equivalent to*

$$P_i q \equiv p_i^{j_i} \pmod{T_i}, \qquad P_i = P/p_i.$$

They are compatible if and only if the corresponding residues agree modulo every $\gcd(T_i, T_j)$ *and with the prescribed class modulo* 5*. When compatible, the last prime lies in one reduced progression*

$$q \equiv r \pmod M, \qquad M = \operatorname{lcm}(5, T_1, \ldots, T_{s-1}).$$

Compatibility does not collapse for large s. In fact, compatible prefixes exist at arbitrary length and above arbitrary prime thresholds. Any global collapse must use the final row.

8. Finite fibres

For a compatible prefix, the final row produces a finite fibre.

Theorem 8.1 (Finite-fibre theorem)**.** *Put* $A = P - 1$*. If the final branch is pure, then*

$$q^4 \mid \left|\operatorname{Res}\left(\Phi_5(X), (X-1)^A - 1\right)\right|.$$

If it is twisted, then

$$q^4 \mid \left|\operatorname{Res}\left(\Phi_5(X), X(X-1)^A + 1\right)\right|.$$

In both cases

$$q \leq 2 \cdot 5^{A/4}.$$

Consequently every fixed prefix has a finite and effectively enumerable last-prime fibre.

The finite fibres admit a stronger universal-level decomposition. With

$$U = -175 + 275\frac{\sqrt{5} - 1}{2},$$

the relevant integer is a product of universal pieces

$$\mathrm{N}(\Phi_d(U)).$$

The piece depends only on d, not on the prefix.

Certified computation 8.2 (Seven empty fibres)**.** The following pure fibres are certified empty without an upper bound on q:

$$(13, 37), \quad (17, 113), \quad (13, 277), \quad (23, 167), \quad (7, 823), \quad (83, 347), \quad (463, 1327).$$

For each pair, every non-excluded universal level is factored completely into proved primes; excluded levels are eliminated by an exact root count in the CRT class; all surviving candidates are checked in the original field.

The universal-level factory currently indexes 356 compatible pairs with $p_1, p_2 < 1500$. Its strict local cache contains 37 factored levels. The latest strict run promotes the seventh fibre $(463, 1327)$; the first remaining blocking levels are $104, 144, 168$.

8.1. **A primitive-divisor gate.** The fibre integers $D_h = N(U^h - 1)$ and $E_h = N(VU^h + 1)$ are not Lucas or Lehmer sequences in the sense of Bilu–Hanrot–Voutier [3]. More precisely, they have minimal characteristic polynomial

$$(X - 3125)(X - 1)(X^2 + 625X + 3125),$$

whereas the binary recurrence belongs only to the trace sequences. The natural conjugate pair also fails the Lucas–Lehmer coprimality hypotheses. Thus the universal BHV threshold 30 cannot be imported as a uniform fibre cutoff. A future primitive-divisor argument would additionally have to control residue degree and the moving CRT class of every possible prime divisor, not merely guarantee the existence of one primitive divisor.

9. Two quantitative walls

The project leaves two distinct obstacles.

9.1. **The local wall.** H4 is a statement about split primes and simultaneous moving-level order conditions. It has density zero unconditionally and the GRH bound of Theorem 5.3, but its vacuity is open.

Two further gates delimit common approaches. The Kummer tower generated by $\sqrt{5}$ and ε records the 2-primary defects exactly, but H4 selects the variable level $v_2(p-1)$ and retains an independent odd-order premise. It is therefore not a lifting problem for one fixed prime along a single $\mathbb{Z}_2$-extension, and the Iwasawa invariants λ, μ alone do not imply a cutoff.

At the unique place $\mathfrak{q} \mid 2$ of $\mathbb{Q}(\sqrt{5})$, the extended logarithms satisfy

$$v_{\mathfrak{q}}(\log_2 \varepsilon) = 1, \qquad v_{\mathfrak{q}}(\log_2 5) = 2.$$

For the two natural forms associated with an H-profile,

$$\Lambda_n^{\pm} = (n-1)\log_2 5 \pm 2n \log_2 \varepsilon,$$

one has

$$\Lambda_n^{\pm} \neq 0, \qquad v_{\mathfrak{q}}(\Lambda_n^{\pm}) \leq 3.$$

Thus the profile does not furnish a deep form in the two fixed 2-adic logarithms to which a bound of Yu [11] could be applied.

9.2. **The global wall.** Even under H4, one must prove that no compatible prefix has a surviving final prime. Finiteness fibre by fibre does not imply that the union over infinitely many prefixes is empty.

There is, however, a simple global shadow.

Theorem 9.1 (Fermat shadow)**.** *Let n be squarefree, $5 \nmid n$, and suppose that it satisfies Agrawal's congruence at $r = 5$. Then*

$$\boxed{5^{n-1} \equiv 1 \pmod{n}.}$$

Thus every composite such n is a Fermat pseudoprime to base 5.

Proof. For $p \mid n$, put $m = n/p$. The local norm condition gives $5^{m-1} \equiv 1 \pmod{p}$, while Fermat gives $5^{p-1} \equiv 1 \pmod{p}$. Since

$$n - 1 = p(m-1) + (p-1),$$

we have $5^{n-1} \equiv 1 \pmod{p}$. Squarefreeness and CRT finish the proof. □

If $M_s(X)$ counts squarefree counterexamples with s factors and $P_5(X)$ counts base-5 Fermat pseudoprimes, then

$$M_s(X) \le P_5(X), \qquad \sum_s M_s(X) \le P_5(X).$$

Li and Pomerance give a modern upper bound for $P_a(X)$ uniform in the base and without a coprimality hypothesis [8]. There is an absolute X_0 such that, uniformly for $2 \le a \le X$ and $X \ge X_0$,

$$P_a(X) \le X^{1 - \frac{\log\log\log X}{2\log\log X}}.$$

Hence

$$\boxed{\sum_{s \ge 2} M_s(X) \le X^{1 - \frac{\log\log\log X}{2\log\log X}}}$$

for sufficiently large X. This is an unconditional sparsity theorem, not a vacuity theorem. Pomerance's classical $XL(X)^{-1/2}$ theorem [10] is cited only for historical comparison.

9.3. **An aureo–Korselt criterion.** Assume H4. For a squarefree composite n, write $\nu = n \bmod 5$. For $p \mid n$, put

$$j_p = \begin{cases} 0, & p \equiv \nu \pmod{5}, \\ 2, & p \not\equiv \nu \pmod{5}, \end{cases}$$

and let T_p be the order of $\zeta_5 - 1$ in $\mathbb{F}_{p^4}^{\times}$. Then n is a counterexample if and only if

$$\omega(n) \ge 3 \text{ is odd}, \quad p \equiv \pm 2 \pmod{5}\ (p \mid n), \quad \nu \in \{2, 3\},$$

and

$$\boxed{T_p \mid n/p - p^{j_p} \qquad (p \mid n).}$$

This criterion assembles the generalized CRT rows and the final finite fibre in one exact statement. It is analogous in role, but not identical, to Korselt's criterion: in particular it does not assert $p - 1 \mid n - 1$. Consequently the infinitude theorem of Alford–Granville–Pomerance and Heath–Brown's upper bound for three-factor Carmichael numbers do not, by themselves, prove that the aureo–Korselt system is empty [2, 6].

10. Certified searches and statistical interpretation

The principal finite datasets are:

(1) 20,000 indices $n \equiv 4 \pmod 5$, $n \leq 100000$: 9,725 nontrivial H_n, 4,522 distinct prime factors, and zero split factors, with proved factorizations.
(2) All 25,422,719 split primes below 10^9: zero complete H-profiles and three complete J-profiles.
(3) The three J-profiles lie exactly on the complementary-factorization line $\mathrm{lcm}(R, E) = (p-1)/2$.
(4) The absence of an H-profile below 10^9 is compatible with the calibrated random-order model. It is evidence, not a proof of H4.
(5) In the three-factor search, no counterexample occurs in the declared finite boxes. The loop bounds are not theoretical thresholds.

Every certificate records the detector criterion, exact reconstruction, primality evidence, original-field checks, and SHA-256 hashes. A timeout is classified as open, never as an empty fibre.

11. Formalization

The standalone Lean development consists of eight modules, 646 source lines, and 45 named declarations. It contains no `sorry`, `admit`, or added axioms. It formalizes:

(1) the index lemma in multiplication form;
(2) the parity core of the J-inertia theorem;
(3) the arithmetic and exponent core of the canonical inert support;
(4) the golden-cyclotomic entanglement identity as a universal polynomial identity;
(5) the Fibonacci–Lucas bridge for the inert support statements;
(6) quadratic-reciprocity specialization for the nonsquareness of 5;
(7) the squarefree arithmetic glue proving the Fermat shadow from its local norm hypotheses;
(8) paper-level inert J_n corollaries with hypotheses stated modulo 5.

12. Open problems

(1) Prove or refute H4.
(2) Decide whether the union of finite fibres is empty for $s = 3$, and for all odd $s \geq 5$.
(3) Extend the universal-level factory far enough to certify a substantial initial segment of compatible prefixes.
(4) Develop a primitive-prime-ideal theorem with residue-degree and ray-class control for the fourth-order fibre sequences.
(5) Develop a horizontal Kummer sieve in which the 2-adic depth varies with p, rather than a fixed-prime Iwasawa specialization.
(6) Determine whether the GRH upper bound for H-profiles has a matching order of magnitude or whether H4 is genuinely vacuous.

Appendix A. Status table

Statement	Status	Dependency
Local semigroup/group structure	theorem	elementary algebra
Golden moment $M_2 = 3\,\mathrm{ind}_5(\varepsilon)$	theorem	local calculation
Quartic rigidity $S = \langle p \rangle$	theorem	all local rows

H4 support equivalence	theorem of equivalence	no vacuity claim
Density zero of H candidates	theorem	finite Chebotarev
$N_H(X) \ll X/(\log X)^2$	theorem under GRH	Kummer composita
Tail density $\Delta = 1 - 40\mathcal{A}/19$	theorem under GRH	UKS proved under GRH
Squarefree CRT and finite fibres	conditional reduction	H4 for skeleton
Seven listed fibres empty	certified	exact factorization/sieve
Fermat shadow $5^{n-1} = 1 \bmod n$	theorem	norm plus CRT
Li–Pomerance bound for $\sum_s M_s(X)$	external theorem	uniform base-5 bound
BHV threshold for D_h, E_h	inapplicable	sequences have order 4
Iwasawa/Yu attacks	delimitations	H4 remains open
Full squarefree Agrawal conjecture at $r = 5$	open	H4 and fibre vacuity

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